Network Influence Analysis Maximization Framework

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1 Abstract

To create a framework for the analysis of influence within networks, particularly the investigation of heuristics for finding the most influential nodes in an efficient manner.

2 Problem Discussion

Network Influence Analysis relates to studying social networks to determine the set of k nodes which have the most 'influence', or greatest spread of information through the network, in the network. A brute force approach may have a running time on the order of the factorial of the number or nodes in a network, using certain heuristics such as the degree of a node to choose the seed nodes is significantly more efficient running in polynomial time, I will focus on allowing for simple implementations and analysis of different heuristics.

There is a significant amount of information out there regarding how these models are implemented and computed, there are graph packages in most programming languages which make it relatively simple to implement them as described.

A typical piece of research in this area would be to attempt to find a new heuristic to find the set of seed nodes which have a greater influence than other heuristics. To do this the researcher would need to implement the model, implement representing their chosen networks in their model, implement the known heuristics and their new hypothesised heuristic and analyse the influence spread of each of them to be able to see if there is a measurable difference.

Creation of a framework for performing these tasks would allow for creating such a study to be simpler, quicker and more reproducible due to a consistent environment.

3 Objectives

- Facilitate import and export of different graph formats
- Generation of synthetic graphs (**Kleinberg, Eppstein**)
- Running of the Independent Cascade Model (**IC**)
- Running of the Linear Threshold Model (**LTM**)
- Running of the Decreasing Cascade Model (**DC**)
- Different Sampling of large graphs to make them manageable to perform computation on
- Implement different heuristics to find seed nodes
- Allow simple creation of new sampling methods

- Allow simple creation of new heuristics
- Python **whl** and **tarball** packaging to allow easy installation on various systems

4 Methods and Methodology

The project will be broken up into three main stages; research, implementation and evaluation. During each stage I will be in contact with my supervisor to ensure that I am on the right track and to keep me accountable.

The research stage will require a detailed literature review of articles and papers pertaining to network influence modelling, methods, heuristics and sampling. I will identify the validity and applications of each source, find if there are any gaps in the presented knowledge to research further. I will repeat this process until my supervisor and I are happy the full scope of the project has been thoroughly reviewed.

The implementation stage of the project will require a methodology that can be adapted to changing requirements or a shift in focus based on information learnt in the later stages of the research phase. I will use an agile-like methodology, of one to two week sprints determined by meetings with my supervisor with any changes to requirements determined during these meetings with future sprints based on the previous.

The final stage will be to use my framework to replicate the results of the researched papers, showcase how to create new sampling methods and heuristics and how to compare the results of different sampling, heuristics and models on different networks.

5 Timeline

The following is based on the university week schedule with week 0 representing freshers' week and week 15 representing the start of the second term. Bold text denotes a deliverable that will be submitted as part of the assessment of this project.

5.1 Deliverables

1. **Specification** - Week 0 to Week 2
2. **Progress Report** - Week 2 to Week 17
3. **Presentation and Demonstration** - Week 17 to Week 28
4. **Final Report** - Week 17 to Week 25

5.2 Milestones

1. **Literature Review** Weeks 0 to Week 2
2. **implementation** Weeks 2 to 13
3. **Replication** Weeks 13 to 17
4. **Documentation and Reporting** Weeks 17 to 23
5. **Polishing up** Weeks 23 to 25

6 Resources and Risks

In terms of non standard tools and programs my project will include some open source large network datasets as model data. If these are to become unavailable from one source not only are they saved in the repository under version control they are also available from multiple sources online.

My code and all documentation is stored in version control locally and with a remote backup in the form of GitHub. I will be using Python3 (Version 3.9 or above), within my program I will use some common libraries such as **NetworkX** for the implementation of the program, Documentation will be generated from docstrings and automatically created into web pages which will be easily searchable with internal links between all relevant sections.

7 Legal, Social, Ethical and Professional Considerations

At this stage there are no Legal, Social, Ethical or Professional considerations to take into account.

The project does not require collection of any user data, it is not going to be processing user data.